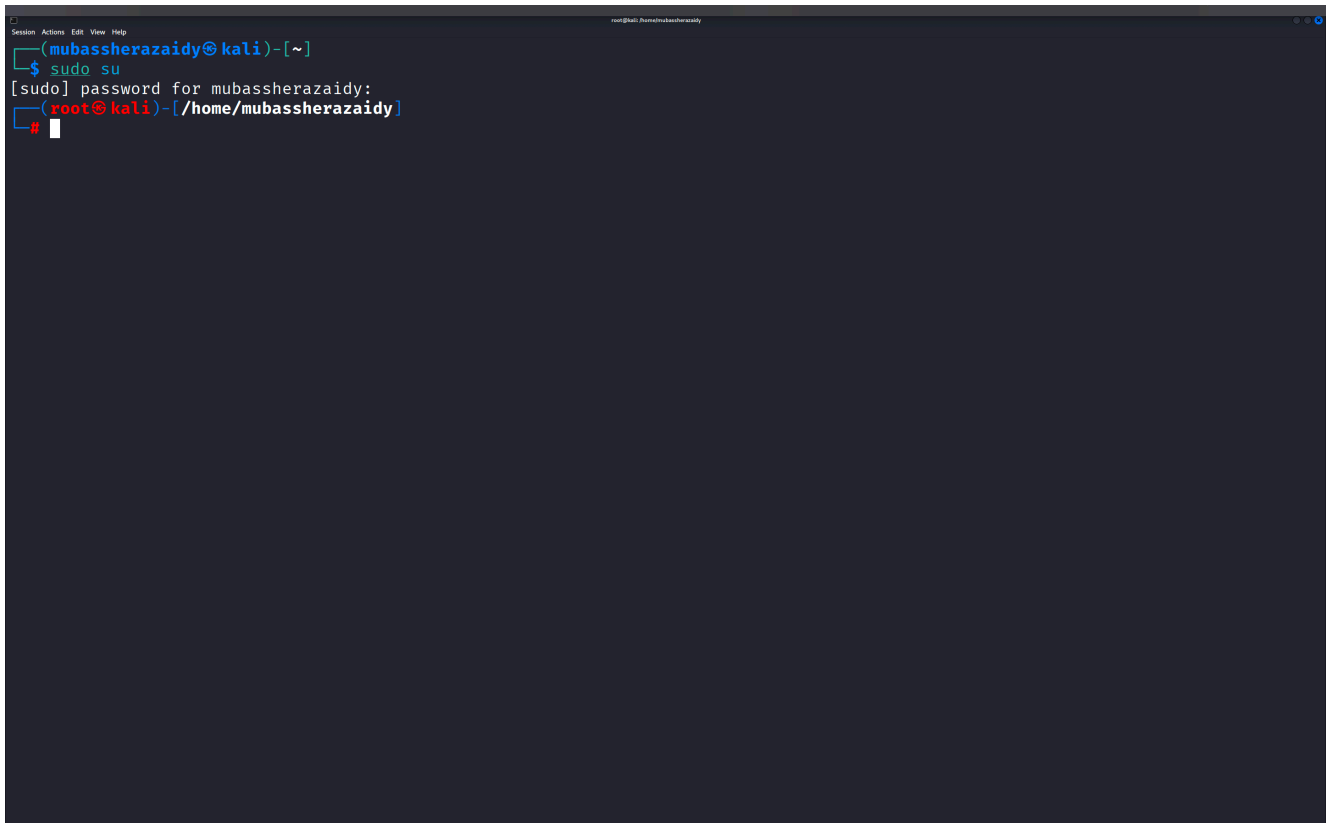


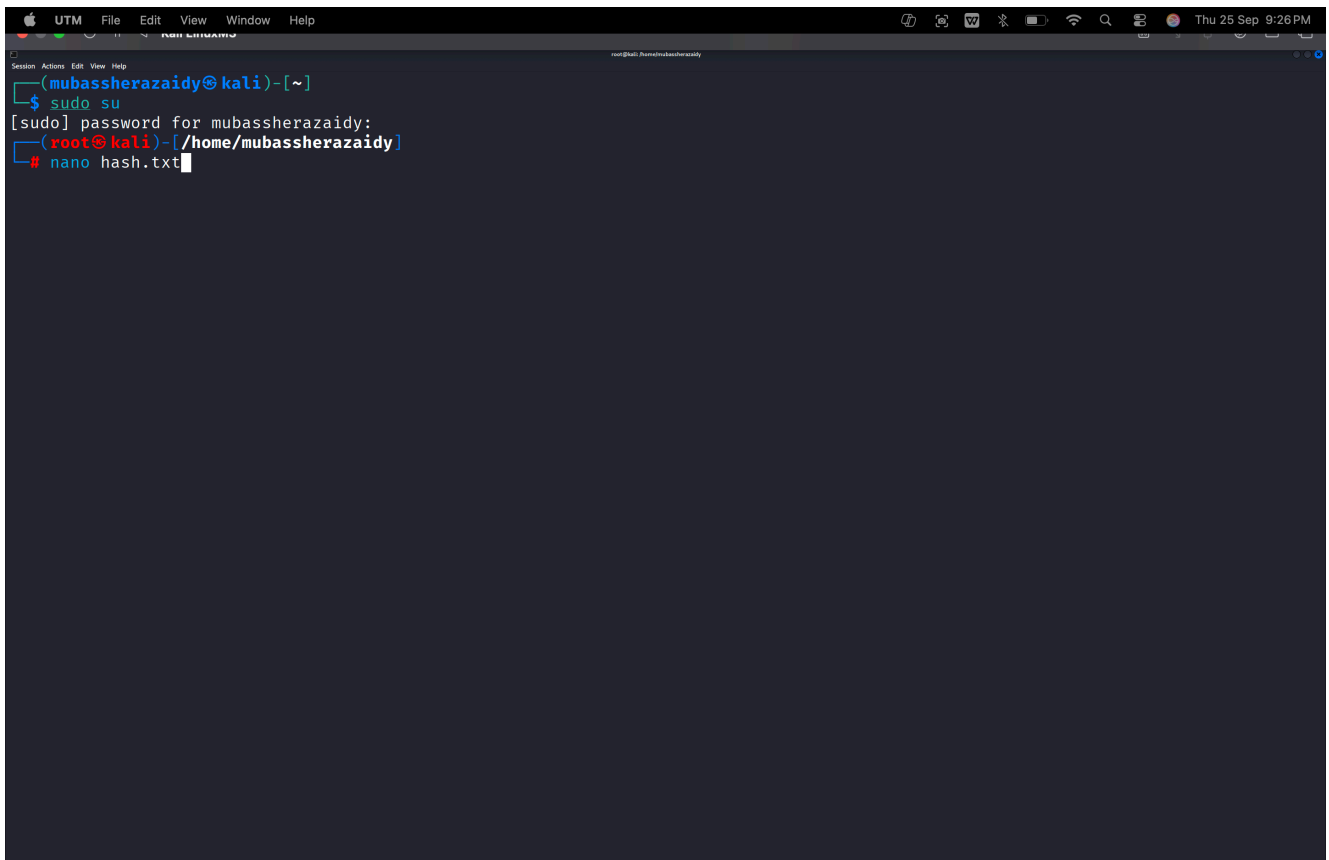
# **HASH CRACKING USING HASHCAT TOOL**

STEP 1 : Open the Kali Linux terminal and run command `sudo su` to switch to the root user.



```
Session Actions Edit View Help
root@kali: ~
(mubassherazaidy@kali)-[~]
$ sudo su
[sudo] password for mubassherazaidy:
(mubassherazaidy@kali)-[~]
#
```

STEP 2: Create a new text file using the nano editor



The screenshot shows a terminal window within a UTM virtual machine. The terminal has a dark background with light blue and red text. The prompt is `(mubassherazaidy@kali)~`. The user enters `sudo su`, and the system prompts for the password for `mubassherazaidy`. After the password is entered, the prompt changes to `(root@kali)~`. The user then enters `nano hash.txt`, which opens the nano text editor. The nano editor interface is visible, showing the file name `hash.txt` and the current line number `1`. The editor is in insert mode, indicated by the `^_` symbol. The terminal window has a menu bar with `Session`, `Actions`, `Edit`, `View`, and `Help`. The top of the window shows the UTM application menu and system status icons.

```
UTM File Edit View Window Help
root@kali:~# mubassherazaidy
Session Actions Edit View Help
(mubassherazaidy@kali)~]
$ sudo su
[sudo] password for mubassherazaidy:
(root@kali)~]
# nano hash.txt
```

STEP 3 : In the new txt file save the hash code you want to crack and save it by pressing `ctrl+X+Y` and `Enter`



The image shows a terminal window with the GNU nano 8.6 text editor. The editor is editing a file named `hash.txt`. The first line of the file contains the hash code `48bb6e862e54f2a795ffc4e541caed4d`. The bottom status bar displays various keyboard shortcuts for editing and navigation.

```
GNU nano 8.6 hash.txt
48bb6e862e54f2a795ffc4e541caed4d

[ Read 1 line ]
^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo      M-A Set Mark
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^/ Go To Line M-E Redo      M-6 Copy
```

STEP 4: Identify the hash format using the `hashid` command and paste the hash you want to crack .

```
mubassherazaidy@kali:~$ sudo su
[sudo] password for mubassherazaidy:
(root@kali)~# nano hash.txt

(root@kali)~# hashid 48bb6e862e54f2a795ffc4e541caed4d
Analyzing '48bb6e862e54f2a795ffc4e541caed4d'
[+] MD2
[+] MD5
[+] MD4
[+] Double MD5
[+] LM
[+] RIPEMD-128
[+] Haval-128
[+] Tiger-128
[+] Skein-256(128)
[+] Skein-512(128)
[+] Lotus Notes/Domino 5
[+] Skype
[+] Snefru-128
[+] NTLM
[+] Domain Cached Credentials
[+] Domain Cached Credentials 2
[+] DNSSEC(NSEC3)
[+] RAdmin v2.x

(root@kali)~#
```

STEP 5 : After identifying the format we can execute the hashcat command to initiate the cracking process.

The hash format is MD5

```
mubassherazaidy@kali:~$ sudo su
[sudo] password for mubassherazaidy:
(mubassherazaidy@kali)~$ nano hash.txt
(mubassherazaidy@kali)~$ hashid 48bb6e862e54f2a795ffc4e541caed4d
Analyzing '48bb6e862e54f2a795ffc4e541caed4d'
[+] MD2
[+] MD5
[+] MD4
[+] Double MD5
[+] LM
[+] RIPEMD-128
[+] Haval-128
[+] Tiger-128
[+] Skein-256(128)
[+] Skein-512(128)
[+] Lotus Notes/Domino 5
[+] Skype
[+] Snefru-128
[+] NTLM
[+] Domain Cached Credentials
[+] Domain Cached Credentials 2
[+] DNSSEC(NSEC3)
[+] RAdmin v2.x
(mubassherazaidy@kali)~$
```

STEP 6: Execute the command `hashcat -m 0 -a 0 hash.txt /usr/share/wordlists/rockyou.txt --show` to begin the cracking .

Where: `-m 0` : specifies the hash type (MD5)

`-a 0` : sets the attack mode to dictionary-based.

`Hash.txt` : File in which the hash is saved

`/usr/share/wordlists/rockyou.txt` - The path to the wordlist file used for the attack.

```
Session Actions Edit View Help
root@kali:~# cat hash.txt
Analyzing '48bb6e862e54f2a795ffc4e541caed4d'
[+] MD2
[+] MD5
[+] MD4
[+] Double MD5
[+] LM
[+] RIPEMD-128
[+] Haval-128
[+] Tiger-128
[+] Skein-256(128)
[+] Skein-512(128)
[+] Lotus Notes/Domino 5
[+] Skype
[+] Snefru-128
[+] NTLM
[+] Domain Cached Credentials
[+] Domain Cached Credentials 2
[+] DNSSEC(NSEC3)
[+] RAdmin v2.x
--End of file 'hash.txt'--

(root@kali)~/home/mubassherazaidy
# hashcat -m 0 -a 0 hash.txt /usr/share/wordlists/rockyou.txt
hashcat (v6.2.6) starting

OpenCL API (OpenCL 3.0 PoCL 6.0+debian Linux, None+Asserts, RELOC, SPIR-V, LLVM 18.1.8, SLEEF, POCL_DEBUG) - Platform #1 [The poc

* Device #1: cpu--0x000, 2557/5178 MB (1024 MB allocatable), 6MCU

Minimum password length supported by kernel: 0
Maximum password length supported by kernel: 256

INFO: All hashes found as potfile and/or empty entries! Use --show to display them.

Started: Thu Sep 25 22:12:40 2025
Stopped: Thu Sep 25 22:12:40 2025

(root@kali)~/home/mubassherazaidy
#
```

STEP 7 : Once the command completes, add `--show` to display the result  
In conclusion, the hash `48bb6e862e54f2a795ffc4e541caed4d` was successfully cracked, and the password is easy.

```
root@kali: ~# hashcat -m 0 hash.txt /usr/share/wordlists/rockyou.txt
Analyzing '48bb6e862e54f2a795ffc4e541caed4d'
[+] MD2
[+] MD5
[+] MD4
[+] Double MD5
[+] LM
[+] RIPEMD-128
[+] Haval-128
[+] Tiger-128
[+] Skein-256(128)
[+] Skein-512(128)
[+] Lotus Notes/Domino 5
[+] Skype
[+] Snefru-128
[+] NTLM
[+] Domain Cached Credentials
[+] Domain Cached Credentials 2
[+] DNSSEC(NSEC3)
[+] RAdmin v2.x

(root@kali)~/home/mubassherazaidy
# hashcat -m 0 hash.txt /usr/share/wordlists/rockyou.txt

hashcat (v6.2.6) starting

OpenCL API (OpenCL 3.0 PoCL 6.0+debian Linux, None+Asserts, RELOC, SPIR-V, LLVM 18.1.8, SLEEF, POCL_DEBUG) - Platform #1 [The pocl project]

* Device #1: cpu--0x000, 2557/5178 MB (1024 MB allocatable), 6MCU

Minimum password length supported by kernel: 0
Maximum password length supported by kernel: 256

INFO: All hashes found as potfile and/or empty entries! Use --show to display them.

Started: Thu Sep 25 22:19:58 2025
Stopped: Thu Sep 25 22:19:58 2025

(root@kali)~/home/mubassherazaidy
# hashcat -m 0 hash.txt /usr/share/wordlists/rockyou.txt --show
48bb6e862e54f2a795ffc4e541caed4d:easy

(root@kali)~/home/mubassherazaidy
#
```